Module-1 CS Introduction

Que-1. what is meaning of cyber security.

Ans. Cybersecurity is the discipline of protecting digital assets—systems, networks, devices, and data—from unauthorized access, attacks, damage, or disruption.

- Cybersecurity is a strategic framework of technologies, processes, and controls designed to manage risk, safeguard information integrity, ensure confidentiality, and maintain availability of digital systems.

Que-2. what are the main objectives of cyber security?

Ans. Confidentiality — “Only the right eyes see the data.”

This is about preventing unauthorized access.

Corporate reality check:
If sensitive data leaks, trust evaporates, lawsuits appear, and careers end.

How it’s achieved:

- Encryption
- Authentication (passwords, biometrics, MFA)
- Access controls

Poetic truth:

Secrets are currency. Cybersecurity makes sure they aren’t counterfeit or stolen.

Integrity — *“Data must stay pure.”*

This ensures information is accurate, complete, and unaltered.

Why it matters:
Imagine financial records quietly edited or flight data tampered with. Chaos. Literal chaos.

How it’s protected:

- Hashing
- Digital signatures
- Version control & audit logs

No sugar-coating:
Corrupted data is worse than stolen data—because you might trust a lie.

Availability — *“Systems must work when needed.”*

Data and services should be accessible on demand.

Business impact:
Downtime = lost revenue, broken operations, angry customers.

Defended against:

- DDoS attacks
- Hardware failures
- Ransomware

Old-school logic still wins here:
If the system is down, security is irrelevant.

Que-3. What is offensive and defensive in cyber security?

Ans. Offensive security is about thinking like the attacker to break systems *before* real criminals do.

Defensive security is about protecting, monitoring, and responding to attacks in real time.

Que-4. what is cyberspace and low.

Ans. Cyberspace is the virtual environment created by interconnected digital systems where online activities happen. It's the invisible digital world where data lives, moves, and gets attacked.

Cyber law is the legal framework that governs activities, rights, and crimes in cyberspace. Cyber law deals with legal issues related to the use of computers, networks, data, and digital communication.

Que-5. What is cyber welfare?

Ans. Cyber welfare refers to protecting, supporting, and promoting the well-being of individuals and society in the digital world. It's about making cyberspace safe, ethical, inclusive, and humane.

Not just secure systems—but secure people.

Que-6. Explain the Types of Hacker

Ans. Types of Hackers

White Hat Hackers (Ethical Hackers)

The good guys. Legally authorized. Professionally dangerous—in a good way.

What they do:

- Find security vulnerabilities
- Perform penetration testing
- Secure systems before criminals attack

Work with: Companies, governments, defense orgs
Legality: Legal
Motive: Protection, prevention, professionalism

Truth bomb:

White hats break systems to build trust.

Black Hat Hackers

The criminals. No permission. All damage.

What they do:

- Steal data & money
- Spread malware & ransomware
- Exploit systems for personal gain

Legality: Illegal
Motive: Profit, power, chaos

No sugar-coating:

Skill without ethics is just a weapon.

Grey Hat Hackers

The rule-benders. Ethical curiosity, questionable legality.

What they do:

- Hack without permission
- Report vulnerabilities (sometimes)
- May ask for a reward

Legality: Illegal / unclear
Motive: Mixed—learning, exposure, ego

Straight talk:

Good intentions don’t cancel illegal actions.

Script Kiddies

Low skill. High confidence. Dangerous by accident.

What they do:

- Use pre-written tools
- Don’t understand how attacks work

Legality: Illegal
Motive: Fun, fame, chaos

Hard truth:

Tools don’t make you a hacker—understanding does.

Hacktivists

Ideology-driven hackers. Digital protestors.

What they do:

- Deface websites
- Leak data
- Launch DDoS attacks

Motive: Political or social causes
Legality: Mostly illegal

Old-school wisdom:

Activism without law becomes sabotage.

State-Sponsored Hackers

Government-backed cyber warriors.

What they do:

- Cyber espionage
- Infrastructure attacks
- Surveillance

Legality: Legal to the state, illegal internationally
Motive: National interest

Reality check:

Cyber war is quiet—but relentless.

Insider Hackers

The threat already inside the system.

Who they are:

- Employees
- Contractors
- Former staff

Motive: Revenge, money, negligence

Que-7. What is the full form of SOC in cyber security.

Ans. Security Operations Center

Now the real meaning behind the words:

A Security Operations Center (SOC) is a centralized unit where cybersecurity professionals continuously monitor, detect, analyze, and respond to cyber threats.

Corporate reality:

SOC is the nerve center of an organization's cyber defense.

Que-8. What are the Challenges of Cyber Security

Ans. Major Challenges of Cybersecurity

Evolving Threat Landscape

- Hackers never sleep. Malware, ransomware, phishing, zero-day attacks—they change every day.
- What worked yesterday? Useless today.

Reality: You have to adapt faster than the attackers.

Human Factor

- Weak passwords, careless clicks, insider threats, social engineering.
- Most breaches aren't "tech failures"—they're people failures.

Reality: Training users is just as important as firewalls.

Lack of Skilled Professionals

- There's a massive cybersecurity talent gap globally.
- SOC analysts, ethical hackers, threat hunters—not enough people to fill critical roles.

Reality: Even the best systems fail without skilled hands.

Advanced Persistent Threats (APTs)

- Attackers stay hidden inside systems for months, stealing sensitive info quietly.
- Detection is hard, mitigation is complex.